Sourcing list

DIY CNC epoxy-granite mill · October 2026

Parts to buy for the machine in `model/DIYCNC.FCStd` / `docs/report.txt`, sourced on Amazon (US) where it makes sense. Prices and stock were checked on **2026-10-09/10**. "?" means the price was not visible (Amazon blocks automated page reads); check the live listing. Shipping and tax are not included. Off-Amazon picks are marked "not Amazon": they are there because Amazon doesn't carry the part or is much dearer. Rows marked (opt) or (choice) are left out of the totals.

The same list is in `docs/sourcing.csv` for a spreadsheet.

Totals

The list is in three parts, so you can buy and build in that order: **Part 1** is the machine itself without the expensive equipment (castings, steel, rails, ballscrews, chip protection), **Part 2** is the motors, controller, wiring and spindle, and **Part 3** is the enclosure and flood coolant. The machine runs after Part 2 with brush-on cutting fluid; the castings already have the inserts, gutters, drain and sump space for Part 3.

Part	Section	Priced
1	Castings and hardware	$868.20
1	Rails, screws and mechanics	$290.38
1	Chip protection	$196.87
2	Motors and drivers	$232.12
2	Controller and wiring	$198.59
3	Enclosure and flood coolant	$436.05

Part	About
1. Frame	**$1,355.45**
2. Electronics + spindle	**$430.71** + spindle: $250-375 (1.5 kW 110 V kit) or $430-600 (2.2 kW + 120 V step-up VFD)
3. Enclosure + flood coolant	**$436.05** + 2 unpriced (ACM sheet, Flood coolant pump): about $650-700
Total	**about $2.69-3.09k**

Shipping and tax are extra (Raka ships free over $100). The whiting and Bolt Depot ship heavy or separately; buying the whiting and the steel locally saves most of it. Steel-yard and remnant prices are estimates: call a yard with the cut list.

JLCMC and JLCCNC (checked 2026-10-10). JLCMC list prices are about half of Amazon for rails and couplings (e.g. the Y rail pair $27.40), but US orders prepay a 55 % duty plus DHL shipping, so the whole motion order comes to about $250-300 against about $190 here. There is no ready-made SFU1204 (lead 4) kit, and the 1605 screw-end machining for BK12/BF12 is not confirmed. Worth adding the rails and couplers only to a JLC order you are placing anyway. JLCCNC machining of the three aluminium plates with all holes is estimated at $430-850 delivered (90 % duty prepaid on machined aluminium): only worth it for the X carriage if you want its hole pattern exact. Retail cut-to-size shops (e.g. Metal Supermarkets, about $820 for the tube, risers and aluminium) also cost more than the sources here.

Part 1 - Frame: castings and hardware

Item	Spec	Qty	Listing	Price	Line	Notes
Epoxy	~9.9 kg low-viscosity laminating epoxy (8.5 % mix), plus ~10 % waste	1 (3 gal)	Raka Standard 2:1 Epoxy Laminating Kit: 127 Thin resin + 606 Slow hardener, 3 gal (not Amazon)	$172.99	$172.99	About 12.5 kg, $13.8/kg; free shipping (Raka: orders over $100). Cast the test bars first; if the mix needs more than 8.5 %, add the 1.5 qt kit ($32.99). Alt: US Composites 635, 4 gal $245. Avoid deep-pour casting resins and 1:1 tabletop epoxies.
Fines	Whiting (calcium carbonate) 325 mesh, ~11 kg	1 (50 lb)	Clay-King calcium carbonate whiting, 50 lb (not Amazon)	$0.84/lb	$42.00	No crystalline silica (check the SDS), so a basic N95 is enough. A local pottery supplier saves the heavy shipping. Avoid garden lime and talc.
Aggregate	~80 kg: fine sand 10, coarse sand 19, pea gravel 23, 3/8 in gravel 23 kg, plus test bars	~5 bags	Pool filter sand + Quikrete all-purpose sand + Vigoro pea gravel + 3/8 in stone (Home Depot)	~$7/bag	$48.00	Buy locally. Wash and dry fully.
Alumina grit, coarse	Brown fused aluminium oxide #16 (12-20 is fine), ~21 kg: the coarse sand in the beam and uprights	1 (50 lb)	IMS Blasting brown aluminum oxide, 50 lb (pick 16 grit)	$109.99	$109.99	Free shipping. Alt: Kleen Blast AO5016 $90 + shipping (blast media is excluded from its free shipping). About +10 % frame stiffness on its own (analysis/alumina.py).
Alumina grit, fine	Black aluminium oxide 70 grit, ~11 kg: the fine sand in the beam and uprights	1 (25 lb)	Central Pneumatic black aluminum oxide 70 grit, 25 lb (Harbor Freight, item 56704)	$39.99	$39.99	Local pickup. With the #16: about +17 % frame stiffness in all.
Mould release wax	Partall Paste #2	1	Partall Paste #2, 12 oz (Fiberglass Supply)	$20.96	$20.96	3-5 coats. PVA film is optional (Partall #10, $8.36/pint).
Square tube	2x2x0.120 A500: 4x 12 in, 2x 18 in; 1-1/2x1-1/2x0.120: 4x 12 in, 6x 6 in	~7 ft of each	Steel-yard remnants (local)	~$40	$40.00	About 37 lb at $0.75-1.25/lb. Online cut lengths: Speedy Metals $106 + shipping. Plain steel, not galvanized.

Item	Spec	Qty	Listing	Price	Line	Notes
Steel plate and pads	3/8 in: 2x 68x168 mm; 1/4 x 2-1/2 in: ~4 in	offcuts	Steel-yard offcuts (local)	~$15	$15.00	Upright top plates and the X support pads
Rail risers	Cold-rolled 1018 x 1-1/4 in: 3/4 thick 2x 600 mm; 5/8 thick 2x 400 + 2x 300 mm	~4 + 5 ft	Steel yard (local)	~$40	$40.00	About 26 lb at $1-2/lb. Online: Speedy Metals $60 + shipping. Pieced bars are fine (two rail screws per piece).
Aluminium plates	6061 drops: table 3/4 x 10.5 x 13 in, X 5/8 x 7 x 16.5, Z 5/8 x 6.25 x 12.25	~22 lb	eBay "6061 aluminum plate remnant" / local drops	~$85	$85.00	About $2.3-4.5/lb delivered. Midwest MIC-6 cut pieces are $242 if you would rather not hunt. Check flatness; face the table on the machine.
Threaded rod	1/2-13, cut to 4x 293, 2x 211, 2x 108, 2x 358 mm	1x 6 ft + 1x 3 ft	Everbilt 1/2-13 zinc threaded rod (Home Depot)	$14.93 + $7.93	$22.86	6 ft: 358, 358, 293, 293, 293, 102, 102; 3 ft: 293, 213, 213 (hacksaw).
Coupling nuts	1/2-13, 10	10	Everbilt 1/2-13 coupling nut (Home Depot)	$2.47	$24.70	Tractor Supply 3-packs $5.99 are a bit cheaper
Nuts and washers	1/2-13 nuts x16, 1/2 in grade 8 washers x16, 3/8 grade 8 washers x8	4 packs	Everbilt packs (Home Depot)	~$20	$19.86	Nuts 15-pack $4.89 + 4-pack $1.38; 1/2 washers 25-pack $9.75; 3/8 washers 25-pack $3.84
Beam bolts	3/8-16 x 6 in grade 8 hex bolt	8	Bolt Depot, grade 8 yellow zinc	$1.96	$15.68	Shipping extra; a local fastener supplier may have them
Dowel pins	3/8 x 1-1/4 in	1 (10-pack)	Steel dowel pins (Zoro)	$5.79	$5.79	They only locate the castings
Feet	1/2-13 x 2 in hex bolt + jam nut + rubber pad, per leg	4	Hardware store bolts + rubber anti-vibration pads	~$12	$12.00	The bolt screws into the nut in each leg end; turn it to level, lock with the jam nut. Leveling mounts (S&W NYL B24, $9.49 each) if you prefer.
O-ring cord	NBR 3 mm, 2.5 m	1 (3 m)	uxcell nitrile round strip 3 mm x 3 m	$6.56	$6.56	
Cord splice glue	Cyanoacrylate	1	Loctite 406, 20 g	$9.86	$9.86	
M5 cast-in inserts	18 riser hold-downs (+11 for the part 3 enclosure)	1 (30)	uxcell knurled brass M5 x 16 mm	$11.08	$11.08	Screw each to the mould floor with a greased screw so it ends up flush. Cast the enclosure inserts now too.
M6 cast-in inserts	6 (Y BK12/BF12 on the base)	1 (20)	Knurled brass M6 inserts (eBay)	~$16	$16.11	Length not confirmed; 12-16 mm is fine
Low-head screws	DIN 7984 M5 x 20 (8, X risers) and M5 x 25 (10, Y risers)	2 (100)	Zoro Select DIN 7984 low-head, class 8.8	$9.99 + $14.29	$24.28	Head 3.5 mm tall, sits below the rail in the 4.5 mm counterbore
Support screws	M6 x 40 (6, Y BK/BF); M6 x 35 (6, X) + 12 DIN 433 small washers	1 (100) + 3 (2-pack)	Zoro M6x40 100-pack; Home Depot M6x35 2-packs	$10.15 + 3x $3.75	$21.40	Two small washers under each X screw head so it stops short of the beam tube (washers from the hardware store)

Item	Spec	Qty	Listing	Price	Line	Notes
Rail screws	M4 x 16	1 (100)	iexcell M4x16 12.9	$8.26	$8.26	
Block screws	M4 x 20	1 (50)	iexcell M4x20 12.9	$7.86	$7.86	Check length against the plate thickness
PETG filament	~2.7 kg for the mould parts and liners	3 (1 kg)	SUNLU PETG 1 kg	$15.99	$47.97	About 1.5 kg less if the pocket cores are cut from XPS foam instead

Priced subtotal: $868.20

Part 1 - Frame: rails, screws and mechanics

Item	Spec	Qty	Listing	Price	Line	Notes
Y rails + blocks	HGR15 x2 at 600 mm + 4x HGH15CA	1 kit	iMetrx HGR15 kit, 2x 600 mm + 4x HGH15CA	$55.99	$55.99	Used at full length, no cutting. The listing calls the blocks "aluminum", which is probably wrong: check them for play. JLCMC alt: 2x E-BHH15C-2-L600-ZA-C-E30 $27.40 + 55 % US duty prepaid + shipping.
Z rails + blocks	HGR15 x2 at 300 mm + 4x HGH15CA	1 kit	BEVDICNC 2x HGR15-300 + 4x HGH15CA	$35.69	$35.69	Alt: CNCMANS B09N73DY32 $34.98
Y ballscrew kit	SFU1605-550, BK12/BF12, DSG16H	1	Ballscrew kit DSG16H + SFU1605-550 + BK/BF12	$39.39	$39.39	Ships with a 6.35x10 coupler: use the 8x10 below
Z ballscrew kit	SFU1204-300, BK10/BF10, DSG12H	1	CNCYEAH SFU1204 300 mm kit	$32.69	$32.69	Ships with a 6.35x8 coupler: use the 8x8 below
Z motor bracket	NEMA 23 + BK10 (HM10-57)	1	ANWOK NEMA 23 steel bracket, SFU1204	$32.02	$32.02	Same bracket as your X one
Y motor bracket	NEMA 23 + BK12 (HM12-57)	1	ANWOK NEMA 23 steel bracket, SFU1605	$31.36	$31.36	Alt: CNCMANS HM12-57 B09QS2S3BT $28.98
Couplers, Y	8x10 mm, D25 L30	1 (3-pack)	CNCTOPBAOS 8x10 plum coupling, 3 pcs	$13.99	$13.99	Spares; check that D25 fits inside the bracket
Couplers, Z	8x8 mm, D25 L30	1 (2-pack)	Yeebyee 8x8 plum coupling, 2 pcs	$12.89	$12.89	The spare covers X if your kit came with 6.35x8
Grease gun	Mini, 3 oz cartridges	1	Performance Tool W54205 mini grease gun kit	$17.59	$17.59	
Needle tip	For the blocks' M4 nipples	1	LubriMatic 05-037 needle	$6.98	$6.98	

Item	Spec	Qty	Listing	Price	Line	Notes
Grease	Lithium NLGI 2	1 (3x3 oz)	LubriMatic 11312, 3x 3 oz	$11.79	$11.79	
(opt) M4 grease nipples	M4x0.7 straight	1 pack	HARFINGTON 15 pcs M4x0.7	$13.94	$13.94	Only if the block nipples are missing or the coupler won't seat

Priced subtotal: $290.38

Part 1 - Frame: chip protection

Item	Spec	Qty	Listing	Price	Line	Notes
Sump	Full-size steam-table pan, 4 in deep (~13 L)	1	Winco SPJH-104, 22 ga stainless (not Amazon)	$22.16-26.81	$26.81	Not found on Amazon; restaurant-supply stores (KaTom, GoFoodservice $23.63) or a local one. Slides in under the base.
Chip basket	1/3-size pan, 2-1/2 in deep	1	CAC STPT-S25-2 1/3 pan (not Amazon)	$9.98	$9.98	Solid pan: drill 6 mm holes in the floor (or buy a perforated 1/3 x 2-1/2 in pan). Sits at the front, under the drain.
Drain nipple	1/2 NPT brass	1 (2-pack)	Legines brass barb 1/2 NPT x 5/8	$10.99	$10.99	Cast into the base, pointing down into the chip basket. +$7.59 shipping.
Way cover strip	0.1 mm 301 stainless	1	SPCCGG 301 strip 0.1 x 300 x 1000 mm	$54.10	$54.10	Gives all three strips. If it turns out soft, use 304 full-hard B09ZQQJ34Z for the 85 mm strips.
Way cover springs	Constant force, for printed rollers	1 (5-pack)	SUS301 constant-force springs, 5 pcs	$35.99	$35.99	One per narrow strip, two for the wide one
Spiral screw cover	20 mm ID, ~250-400 mm	1	MISONONEIK spiral steel bellows cover	~$47	$47.00	Choose size 20-250-40-28 (the cheapest variant, about $47) or 20-400-40-33; variants $47-116
Drip shield	510 x 57 mm	0	Cut from the ACM sheet offcut	$0	$0.00	Stainless was $57 (Speedy Metals 16 ga 12x24); ACM does the job
Cutting fluid	Brush-on / drip, until flood coolant (part 3)	1	Tap Magic or similar cutting fluid	~$12	$12.00	Brush or drip it on; the base gutters and drain still carry it to the pan

Priced subtotal: $196.87

Part 2 - Electronics: motors and drivers

Item	Spec	Qty	Listing	Price	Line	Notes
X/Y closed-loop kits	23HS30-5004D-E1000 2.0 Nm + CL57T	2	STEPPERONLINE closed-loop kit 2.0 Nm + CL57T	$81.00	$162.00	Exact model. Low stock when checked.

Item	Spec	Qty	Listing	Price	Line	Notes
Z driver (owned motor)	DM542T	1	STEPPERONLINE DM542T	$32.99	$32.99	Check that your motor is 4.2 A or less
Motor power supply	48 V, 3 drivers	1	Mean Well LRS-350-48 (7.3 A)	$37.13	$37.13	Trim to about 46 V: CL57T max input is 48 V

Priced subtotal: $232.12

Part 2 - Electronics: controller and wiring

Item	Spec	Qty	Listing	Price	Line	Notes
Controller	FluidNC, 6 axes, external drivers, RS485, 0-10 V	1	Elecrow Corgi ESP32 FluidNC controller (not Amazon)	$49.00	$49.00	Runs gSender or the web UI. Drives an RS485 VFD directly, so no 0-10 V module is needed for one.
Limit switches	Roller lever, 4 needed	1 (5-pack)	Yusheng ME-8108 limit switch set of 5	$16.99	$16.99	Inductive LJ12A3-4-Z/BX (about $4.50 each) also work; power them from 12 V.
E-stop	22 mm mushroom, latching, NC, in a box	1	Generic 22 mm E-stop station (eBay)	$22.39	$22.39	Wire it to cut drive enable and VFD run in hardware. Check it has an NC contact.
Mains inlet	IEC C14 with switch + fuse	1	IEC C14 inlet module (Walmart)	$7.99	$7.99	Often ships without the fuse: buy 5x20 mm fuses.
Logic power supply	24 V, 6.5 A	1	Mean Well LRS-150-24 (TRC Electronics)	$20.39	$20.39	Jameco about $18.80
Electrical box	ABS IP65, 11.8 x 9.8 x 4.7 in	1	LeMotech ABS IP65 junction box 11.8x9.8x4.7	$25.99	$25.99	Holds the controller, three drivers and both supplies on a plywood or aluminium backplate; the VFD mounts outside. Steel alt: VEVOR 12x10x6 $48.90.
Motor, sensor and spindle cable	4-conductor shielded, ~45 ft	45 ft	BuildYourCNC 4-conductor shielded cable	$0.49/ft	$22.05	One spool for the steppers, switches and spindle: 18/4 shielded is enough for a 1.5-2.2 kW spindle (BuildYourCNC). Shield to earth at the VFD end. Check the gauge and that it is stranded for the cable chain.
DIN rail + terminals	UK2.5B kit	1	Erayco 20x UK2.5B + rail	$19.99	$19.99	May need more rail
Cable glands	PG7-PG16	1 (25 pcs)	Nylon cable gland kit (eBay)	$13.80	$13.80	
(opt) Tool setter	NC, magnetic base	1	Z touch tool setter	~$31.61	$31.61	Nice to have; add later

Priced subtotal: $198.59

Part 2 - Electronics: spindle (pick one)

Item	Spec	Qty	Listing	Price	Line	Notes
(choice A) 1.5 kW 110 V water-cooled kit	Spindle + 110 V VFD + 80 mm clamp + pump, ER11/ER16	1	SainSmart / VXB / Huanyang-type kits	$250-375	$300.00	Runs on a normal 120 V outlet. SainSmart $349 (sold out; open-box $249), VXB $374.88, eBay import ~$341 shipped. Check Amazon kits by hand.
(choice B) 2.2 kW 220 V kit + 120 V step-up VFD	ER20 kit + 110-120 V in / 220 V out VFD	1 + 1	RATTMMOTOR 2.2 kW kit + ATO or AT5-2200X VFD	$251 + $178-346	$500.00	Only about 1.2-1.6 kW on a 120 V circuit; full 2.2 kW once you have 240 V (then use the kit's own VFD).
(choice) ER collets	ER11/ER16 or ER20 set to match	1	Chiloskit ER20 13 pcs (or an ER11 set)	$25.21	$25.21	Skip if the kit includes a set
(choice) Spindle pump relay	SSR, DC in, AC out	1	SSR-25DA with heatsink (eBay)	$17.44	$17.44	Runs the spindle water pump whenever the spindle runs. Fuse the branch.
(choice) Spindle coolant	Distilled water + 10-20 % RV antifreeze, bucket	1	Local store	~$10	$10.00	Water-cooled spindles only

Priced subtotal: $0.00

Part 3 - Enclosure and flood coolant

Item	Spec	Qty	Listing	Price	Line	Notes
2020 extrusion	4x766, 2x756, 4x684, 2x600, 3x560 mm	2 (10x 1000 mm)	KathyLiely 2020 T-slot 1000 mm, 10 pcs	$49.99	$99.98	One piece per stick; 5 sticks spare. Take the 2x 244 from your 400s.
ACM sheet	3 mm, all four wall/roof panels	1 (4x8 ft)	Sign supply / Piedmont Plastics (not Amazon)	?	?	The 726x684 sides don't fit any in-stock Amazon sheet. One 4x8 covers all four panels and the drip shield. Amazon 24x36 2-pack (B0F18ZQ7DJ, $65.99) covers back + roof only.
Door	Clear polycarbonate 560x756	1	SIBE polycarbonate 1/8 in, 24x36	$24.63	$24.63	
Corner brackets	~30	1	JCSPBYL 30 sets 2020 corner brackets	$12.99	$12.99	
T-nuts	M5 drop-in, 100	1	POLISI3D 100 M5 drop-in T-nuts	$7.65	$7.65	
Button-head screws	M5 x 10, 100	1	uxcell M5x10 button head, 100	$13.99	$13.99	
Hinges + handle	2020, 6 mm slot	1 kit	KOOTANS 2020 hinge and handle kit	$14.59	$14.59	M4 or M5 not stated

Item	Spec	Qty	Listing	Price	Line	Notes
Gas struts	2x ~50 N, 200 mm	1 pair	SBOTTON 200 mm gas struts, 50 N	$38.76	$38.76	Check force against the door weight
Door catch	Magnetic	1 (4-pack)	Onarway cabinet magnets	$13.99	$13.99	
(opt) Door interlock	2NC key switch	1	CZ-93B safety switch	$22.13	$22.13	Wire into the E-stop or feed-hold loop
Panel trim	U-strip for 3 mm panels	1	BLCCLOY 2020 panel U-strip, 25x 420 mm	$17.99	$17.99	10.5 m; just short of every edge
Brush seal	For 2x 72 mm motor cutouts	1 m	SLZ dust-shoe brush strip 50 mm	$20.00	$20.00	
Bulkhead	1/4 NPT, thin panel	1 (2-pack)	FloFlex 1/4 in bulkhead, short	$9.99	$9.99	Measure the hole size first
Sump filter mesh	100 um	1	100 micron filter sock (cut open)	$10.59	$10.59	Cut open and clamp in the printed filter frame
Magnet bar	100 x 25 mm	1	ULIBERMAGNET 12000 Gs filter rod 100x25	$29.99	$29.99	Lies on the sump floor and catches fine steel
Flood coolant pump	24 V brushless submersible, ~800 L/h, 5 m head, <= 60 mm tall	1	Search Amazon: "24V brushless submersible pump 800L/H 5M"	?	?	Low enough for the 4 in sump and no mains voltage in the coolant. Runs from the 24 V supply via the controller's MOSFET output. Replaces the $113 120 V pump.
Return tubing	8 mm OD PU	1 (5.5 m)	uxcell 8 mm PU tubing	$10.19	$10.19	+$1.21 shipping
Ball valve	1/4 NPT	1	Beduan brass mini ball valve	$7.99	$7.99	
Manifold	1/4 NPT Y	1	FASPARTS brass Y block	$11.99	$11.99	One bullet says 1/2 in thread: check before ordering
Loc-Line nozzles	1/4 in kit	2	Loc-Line LP-40413 coolant hose kit	$15.15	$30.30	
Push-to-connect fittings	8 mm x 1/4 NPT	1 (10)	Qjaiune push fittings	$7.49	$7.49	One at the pump outlet, so the return line unplugs and the sump slides out
Coolant concentrate	Semi-synthetic, 1 gal	1	Fusion Cool 2240	$52.95	$52.95	Mix 5-10 %

Priced subtotal: $436.05 + 2 unpriced: ACM sheet, Flood coolant pump

Already owned

- HGR15-400 rails x2 + 4 HGH15CA blocks (X)
- SFU1204-400 kit with BK10/BF10, DSG12H housing, coupler (X)
- NEMA 23 bracket for BK10 (X)
- Open-loop NEMA 23 (Z)
- Extrusion: 10x 2040 x 400, 4x 2020 x 400, 4x 2020 x 200
- Bambu A1 mini, taps, drill press

Not included

- Cutting tools, a vise and clamps (workholding).
- The water hoses that come with the spindle kit (check the barb size before buying more).
- Wire, ferrules, fuses and breakers for the electrical box.

Check before ordering

1. **Couplers.** Both ballscrew kits ship with 6.35 mm-bore couplers; your motors have 8 mm shafts. Also check your owned open-loop motor's shaft and current.
2. **VFD on 120 V.** No real 2.2 kW drive takes 120 V and outputs 220 V; the step-up drives stop at 1.5 kW. The spindle runs at up to ~1.5 kW now and gets its full 2.2 kW when you switch to the kit's 220 V VFD on 240 V. Use a dedicated 20 A circuit.
3. **No cutting.** Rails, ballscrews and tube are used at the lengths they are sold in; the risers, plates and top plates are ordered cut to size. Only the M12 threaded rod is cut (hacksaw).
4. **Tubes and rebar.** Buy plain (black) steel, not galvanized: epoxy bonds poorly to zinc. Wire-brush off rust and loose scale, degrease just before casting. Stock-length tube is +/-1/8 in on length, with an internal weld seam: file the seam where the spigots and printed sleeves go. 0.120 in wall (11 ga) works as well as 1/8 in.
5. **Imperial sizes.** The plates and risers are already inch sizes in the model. The 3/8 in top plates are 0.5 mm thinner than the modelled 10 mm, which doesn't matter.
6. **Low stock.** Several picks showed "only N left", including the closed-loop kits.